Python for Data Science

Why Python

Python is the most popular language for data science because it is readable, has a huge ecosystem of libraries and works well for everything from quick analysis to production systems.

NumPy and pandas

NumPy provides fast arrays and mathematical operations. pandas builds on it with the DataFrame, a table-like structure for loading, cleaning, filtering, grouping and joining data from CSV files, Excel sheets and databases.

Visualisation

Matplotlib is the foundation for plotting in Python, while seaborn adds statistical charts with sensible defaults. Plotly and Streamlit make it easy to build interactive charts and dashboards.

Machine Learning Libraries

scikit-learn offers a consistent interface for classic machine learning models, preprocessing and evaluation. PyTorch and TensorFlow are the main libraries for deep learning.